

Show understanding of the features provided by a Database Management System (DBMS) that address the issues of a file based approach

9608_s19_qp_13_sc_3.a.i

QUESTION

- 3

A company uses a relational database, `EMPLOYEES`, to store data about its employees and departments.

(a) The company uses a Database Management System (DBMS).

(i) The DBMS has a data dictionary.

Describe what the data dictionary stores.

.....

.....

.....

..... [2]

MARK SCHEME

3(a)(i)	1 mark per bullet point Stores all the information about the database // data about the data // metadata about the data For example, fields, data types, validation, keys	2
---------	---	----------

QUESTION

(ii) Name **and** describe **two** security features provided by a DBMS.

Feature 1

.....

.....

.....

Feature 2

.....

.....

.....

[4]

MARK SCHEME

5(b)(ii)	1 mark for naming, 1 mark for description For example: • Access rights // User accounts • Restrict actions (e.g. read / read-write) of specific users // unauthorised users cannot access the database • Views • Restrict which parts of the database specific users can see • Password // Biometrics // PIN code • Prevents unauthorised access • Automatic Backup • Create regular copies of data in case of loss • Encryption • Data is incomprehensible to unauthorised users	4
----------	--	----------

QUESTION

- (ii) The software company decides to use a database to overcome the limitations of a file-based system. Some of these limitations are addressed through the logical schema.

Name **and** describe **two** levels of the schema of a database.

Name 1

Description

.....

Name 2

Description

.....

[4]

MARK SCHEME

2(b)(ii)	1 mark for each correct name, 1 mark for each matching description, max 2 marks per level External The individual's view(s) of the database Conceptual Describes the data as seen by the applications making use of the DBMS Describes the 'views' which users of the database might have Physical / Internal Describes how the data will be stored on the physical media Logical Describes how the relationships will be implemented in the logical structure of the database	4
----------	--	----------

QUESTION

(b) The database manager is concerned about data integrity.

State what is meant by **data integrity**. Give an example of how the manager can ensure data integrity in the PURPLEGAME database.

.....

.....

.....

.....[2]

MARK SCHEME

2(b)	1 mark for description Ensure data is consistent / accurate // keep data consistent / accurate 1 mark for example from: e.g. - Validation rules - Referential integrity - Verification - Input masks - Setting data types - Removing redundant data - Backup data - Access controls - Audit trail	2
------	--	----------

QUESTION

(b) Name and describe **two** security measures the Network Manager has in place to protect the security of the data held in the DBMS.

1

.....

.....

.....

2

.....

.....

.....

[4]

MARK SCHEME

3(b)	1 Mark for stating the security measure 1 Mark for a corresponding description Maximum 2 marks for each measure Maximum 2 measures Physical measures Locked doors/keyboards etc. Secure methods of access, keypads/ biometric scans etc. Backup of data Regular copies of the data are made If the data is corrupted it can be restored Disk-mirroring All activity is duplicated to a second disk in real time so that if the first disk fails there is a complete copy available Access rights Different access rights for individuals/groups of users To stop users editing data they are not permitted to access By example Encryption If accessed, data cannot be understood by unauthorised personnel Accessed only by those with the decryption key Firewall To stop unauthorised access/hackers gaining access to the computer network Use authentication methods such as passwords and usernames Passwords should be strong / biometrics To prevent unauthorised access to data Anti-malware program To detect / remove / quarantine viruses / key-loggers etc. Carrying out regular scans Concurrent Access Controls // Record locking Closes a record to second user until first update complete To prevent simultaneous updates being lost	4
------	---	---

QUESTION

- 5 (a) A Database Management System (DBMS) provides the following features.

Draw a line to match each feature with its description.

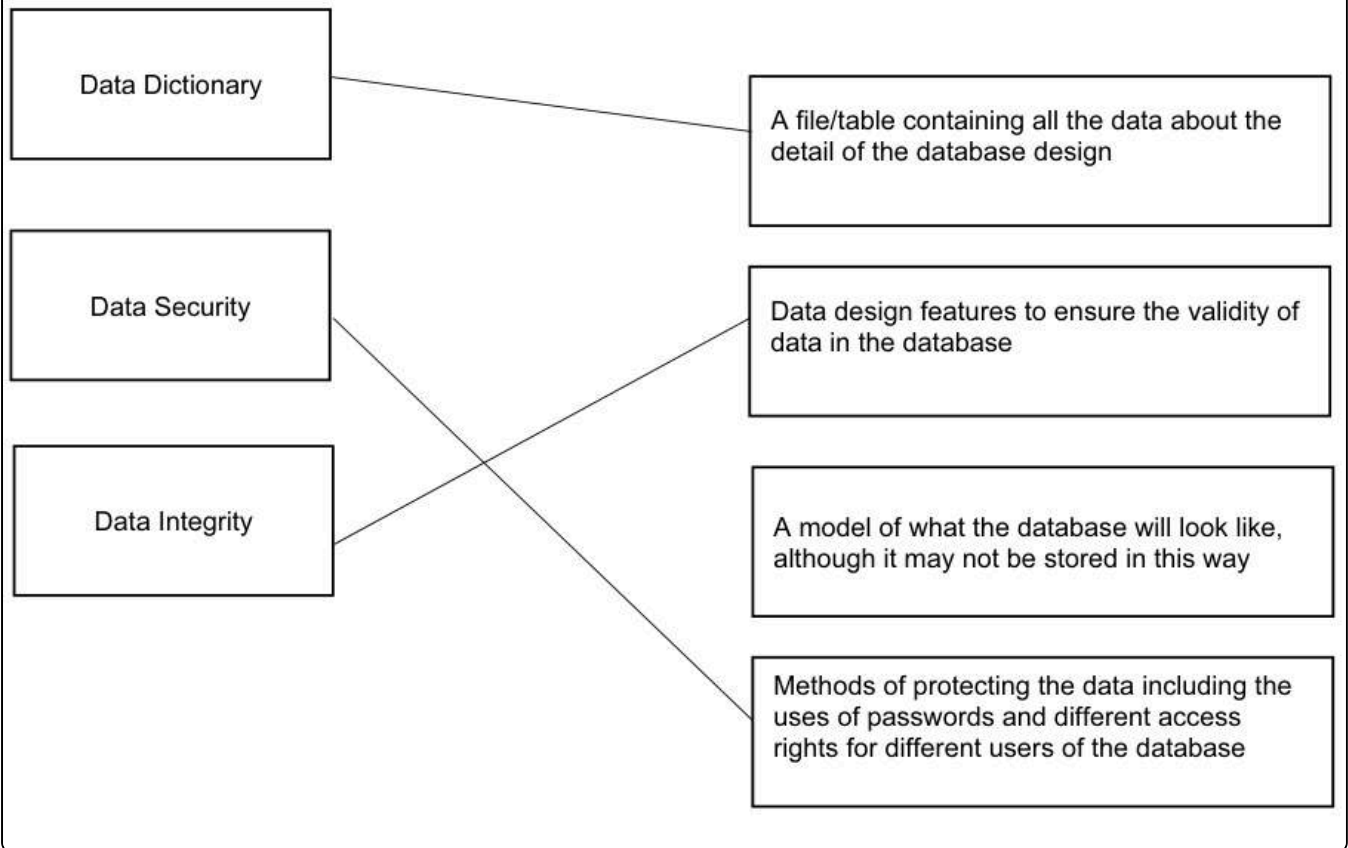
Feature	Description
Data dictionary	A file or table containing all the details of the database design
Data security	Data design features to ensure the validity of data in the database
Data integrity	A model of what the database will look like, although it may not be stored in this way
	Methods of protecting the data including the uses of passwords and different access rights for different users of the database

[3]

MARK SCHEME

5 (a) One mark for each correct line.

[3]



QUESTION

A school stores a large amount of data that includes student attendance, qualification and contact details. The school is setting up a relational database to store these data.

(b) The school needs to safeguard against any data loss.

Describe **three** factors to consider when planning a backup procedure for the data.

Justify your decisions.

- 1
-
-
- 2
-
-
- 3
-
-[6]

MARK SCHEME

(b) One mark for procedure point, one mark for justification.

[6]

Maximum **three** procedures.

- How often should the data be backed up? e.g. at the end of each day
- Justification e.g. student's progress may be edited each day and should not be lost
- What medium should the data be backed up to? e.g. external hard disk drive
- Justification e.g. it has large enough capacity
- Where should the backups be stored? e.g. off-site
- Justification e.g. so if the building is damaged only the original data are lost
- What is backed up? e.g. only updated files ...
- Justification e.g. There are a large number of files and they are not all updated each day
- When should the backup take place? e.g. overnight
- Justification e.g. the system is not likely to be used then
- Who is responsible for performing the backup?
- Justification e.g. otherwise it may not be done
- Make sure the procedure is written down and understood by staff
- Justification e.g. otherwise some data may not be backed up

QUESTION

8 A school stores a large amount of data. This includes student attendance, qualification, and contact details. The school's software uses a file-based approach to store this data.

(a) The school is considering changing to a DBMS.

(i) State what DBMS stands for.

.....[1]

MARK SCHEME

8 (a) (i) Database Management System

[1]

QUESTION

(ii) Describe **two** ways in which the Database Administrator (DBA) could use the DBMS software to ensure the security of the student data.

1

.....

.....

.....

2

.....

.....

.....[4]

MARK SCHEME

- (ii) **One mark** for identifying the way in which the data security is ensured, and **one mark** for a further description.

Maximum of two marks per method. **Maximum of two** methods.

[4]

- Issue usernames and passwords...
 - stops unauthorised access to the data
 - any further expansion e.g. strong passwords / passwords should be changed regularly etc...
- Access rights / privileges...
 - so that only relevant staff / certain usernames can read/edit certain parts of the data
 - can be read only, or full access / read, write and delete
 - any relevant example e.g. only class tutors can edit details of pupils in their tutor group
- Create (regular / scheduled) backups...
 - in case of loss/damage to the live data a copy is available
 - any relevant example e.g. backing up the attendance registers at the end of each day and storing the data off-site/to a separate device
- Encryption of data...
 - if there is unauthorised access to the data it cannot be understood // needs a decryption key
 - any relevant example e.g. personal details of pupils are encrypted before being sent over the Internet to examination boards
- Definition of different views...
 - composed of one or more tables
 - controls the scope of the data accessible to authorised users
 - any relevant example e.g. teachers can only see their classes
- Usage monitoring / logging of activity...
 - creation of an audit /activity log
 - records the use of the data in the database / records operations performed by all users / all access to the data
 - any relevant example, e.g. Track who changed a student's grade

9608_s16_qp_11_sc_8.a.i

QUESTION

- 8** A school stores a large amount of data. This includes student attendance, qualification, and contact details. The school's software uses a file-based approach to store this data.

(a) The school is considering changing to a DBMS.

(i) State what DBMS stands for.

.....[1]

MARK SCHEME

8 (a) (i) Database Management System

[1]

QUESTION

- (ii) Describe **two** ways in which the Database Administrator (DBA) could use the DBMS software to ensure the security of the student data.

1

.....

.....

.....

2

.....

.....

.....[4]

MARK SCHEME

- (ii) **One mark** for identifying the way in which the data security is ensured, and **one mark** for a further description.

Maximum of two marks per method. **Maximum of two** methods.

[4]

- Issue usernames and passwords...
 - stops unauthorised access to the data
 - any further expansion e.g. strong passwords / passwords should be changed regularly etc...
- Access rights / privileges...
 - so that only relevant staff / certain usernames can read/edit certain parts of the data
 - can be read only, or full access / read, write and delete
 - any relevant example e.g. only class tutors can edit details of pupils in their tutor group
- Create (regular / scheduled) backups...
 - in case of loss/damage to the live data a copy is available
 - any relevant example e.g. backing up the attendance registers at the end of each day and storing the data off-site/to a separate device
- Encryption of data...
 - if there is unauthorised access to the data it cannot be understood // needs a decryption key
 - any relevant example e.g. personal details of pupils are encrypted before being sent over the Internet to examination boards
- Definition of different views...
 - composed of one or more tables
 - controls the scope of the data accessible to authorised users
 - any relevant example e.g. teachers can only see their classes
- Usage monitoring / logging of activity...
 - creation of an audit /activity log
 - records the use of the data in the database / records operations performed by all users / all access to the data
 - any relevant example, e.g. Track who changed a student's grade

Show understanding of how software tools found within a DBMS are used in practice

9608_s20_qp_13_sc_6.b

QUESTION

(b) Identify **three** tasks that Sheila can perform using the DBMS developer interface.

- 1
- 2
- 3 [3]

MARK SCHEME

6(b)	1 mark per task to max 3 • Create a table • Set up relationships between tables • Create / design a form • Create / design a report • Create / design a query (NOT run a query)	3
------	--	----------

QUESTION

- (b) The hotel wants to use a Database Management System (DBMS) to set up and manage the database.

Describe, using examples, how the hotel can use the following DBMS tools:

Developer interface

.....

.....

.....

.....

Query processor

.....

.....

.....

.....

[5]

MARK SCHEME

3(b)	1 mark per bullet point to max 2 plus 1 mark for suitable example for each DBMS tool Developer Interface To create user friendly features e.g. forms to enter new bookings To create outputs e.g. report of bookings on a given date To create interactive features e.g. buttons and menus Query processor To create SQL/QBE queries To search for data that meets set criteria, e.g. all bookings for next week To perform calculations on extracted data, e.g. number of empty rooms tomorrow	5
------	--	----------

9608_s19_qp_13_sc_3.a.ii

QUESTION

- (ii) The DBMS has a query processor.

Describe the purpose of a query processor.

.....

.....

.....

..... [2]

MARK SCHEME

3(a)(ii)	1 mark per bullet point to max 2 Allows the user to enter criteria Searches for data which meets the entered criteria Organises the results to be displayed to the user	2
----------	---	----------

9608_s19_qp_12_sc_5.b.iii

QUESTION

- (iii) The DBMS provides software tools for the database developer.

Fill in the names of the missing software tools in the following statements.

A allows a developer to extract data from a database.

A enables a developer to create user-friendly forms and reports.

[2]

MARK SCHEME

5(b)(iii)	1 mark per bullet • Query Processor • Developer Interface	2
-----------	--	----------

QUESTION

(iii) A feature of the DBMS software is a query processor.

Describe how the school secretary could use this software.

.....

.....

.....

.....[2]

MARK SCHEME

(iii) Two points from:

[2]

- Set up search criteria
- To find / retrieve / return the data that matches the criteria
- Any relevant example e.g. find pupils who were absent on a particular day

QUESTION

(iii) A feature of the DBMS software is a query processor.

Describe how the school secretary could use this software.

.....

.....

.....

.....[2]

MARK SCHEME

(iii) Two points from:

[2]

- Set up search criteria
- To find / retrieve / return the data that matches the criteria
- Any relevant example e.g. find pupils who were absent on a particular day